

Grouped frequency tables and histograms



1

Class	Frequency
30 – 39	3
40 – 49	5
50 – 59	3
60 – 69	8
70 – 79	2
80 – 89	4

2

Class	Class centre (x)	Frequency (f)	xf
21 – 25	23	3	69
26 – 30	28	4	112
31 – 35	33	2	66
36 – 40	38	8	304
41 – 45	43	3	129
46 – 50	48	5	240

3

Class	Class centre (x)	Frequency (f)	xf
11 – 15	13	6	78
16 – 20	18	8	144
21 – 25	23	2	46
26 – 30	28	2	56
31 – 35	33	5	165
36 – 40	38	1	38

4



a

Class	Class centre (x)	f	xf
1 – 9	5	8	40
10 – 18	14	6	84
19 – 27	23	4	92
28 – 36	32	6	192
37 – 45	41	8	328
Total		32	736

b

23.00

5 30 – 39

6 23.0

7

a 9.3

b

1 – 5

8

a Ungrouped

b

Grouped

9

a 33.5

b

21

10 $19 < \text{height} \leq 28$

11

a 7

b

4

12

a

Score (x)	Frequency (f)
10 – 19	3
20 – 29	4
30 – 39	2
40 – 49	1
50 – 59	1

b

Score (x)	Frequency (f)
30 – 39	5
40 – 49	4
50 – 59	2
60 – 69	8
70 – 79	3
80 – 89	3



13

a $20 \rightarrow 29, 30 \rightarrow 39, 40 \rightarrow 49, 50 \rightarrow 59$, and $60 \rightarrow 69$

b

Score (x)	$20 \rightarrow 29$	$30 \rightarrow 39$	$40 \rightarrow 49$	$50 \rightarrow 59$	$60 \rightarrow 69$
Frequency (f)	3	2	1	3	2

14

a $10 \rightarrow 19, 20 \rightarrow 29, 30 \rightarrow 39, 40 \rightarrow 49, 50 \rightarrow 59, 60 \rightarrow 69$, and $70 \rightarrow 79$

b

Score (x)	Frequency (f)
$10 \rightarrow 19$	7
$20 \rightarrow 29$	3
$30 \rightarrow 39$	2
$40 \rightarrow 49$	3
$50 \rightarrow 59$	6
$60 \rightarrow 69$	4
$70 \rightarrow 79$	4

15

a

Score (x)	Frequency (f)
$1 - 5$	6
$6 - 10$	12
$11 - 15$	7
$16 - 20$	3
$21 - 25$	9
$26 - 30$	2
$31 - 35$	4

b

Score (x)	Frequency (f)
$20 - 29$	150
$30 - 39$	140
$40 - 49$	160
$50 - 59$	150
$60 - 69$	130
$70 - 79$	110
$80 - 89$	120
$90 - 99$	130

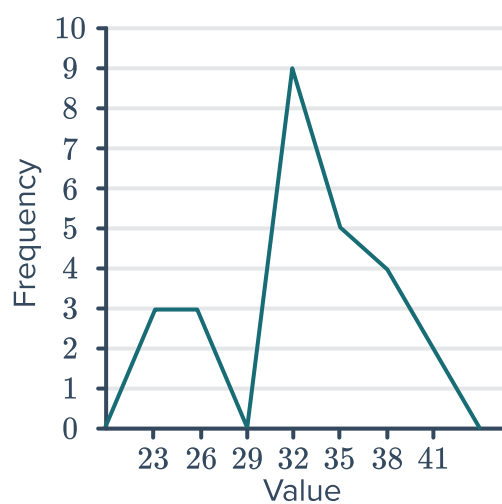
16



a

Class interval	Class centre	Frequency
$21.5 \leq p < 24.5$	23	3
$24.5 \leq p < 27.5$	26	3
$27.5 \leq p < 30.5$	29	0
$30.5 \leq p < 33.5$	32	9
$33.5 \leq p < 36.5$	35	5
$36.5 \leq p < 39.5$	38	4
$39.5 \leq p < 42.5$	41	2

b



c 0

d 17

e 15

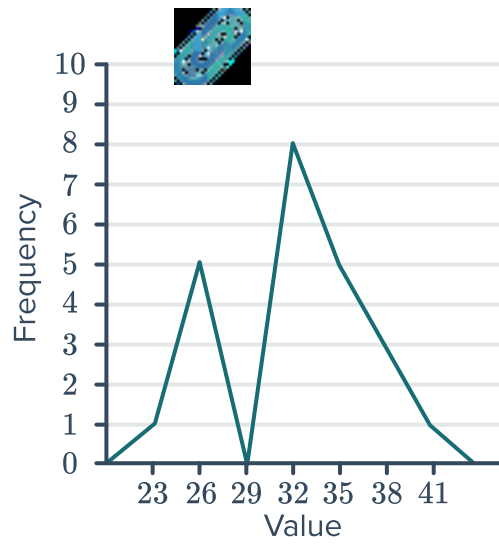
f 20

17

a

Class limits	Midpoint	Frequency
21.5 – 24.5	23	1
24.5 – 27.5	26	5
27.5 – 30.5	29	0
30.5 – 33.5	32	8
33.5 – 36.5	35	5
36.5 – 39.5	38	3
39.5 – 42.5	41	1

b



c 0

d 18

e 14

f 17

Applications

18

a Yes

b Yes

c Yes

d No

19

a

Heart Rate	Class Centre	Frequency
45 – 54	49.5	15
55 – 64	59.5	25
65 – 74	69.5	26
75 – 84	79.5	30

b 96 people

c 51 people

d 40 people

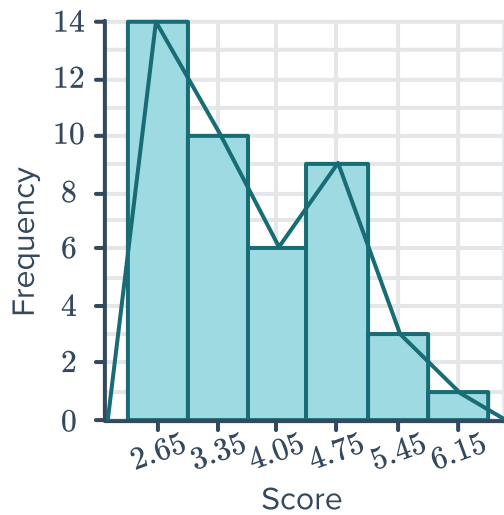
20

a



b

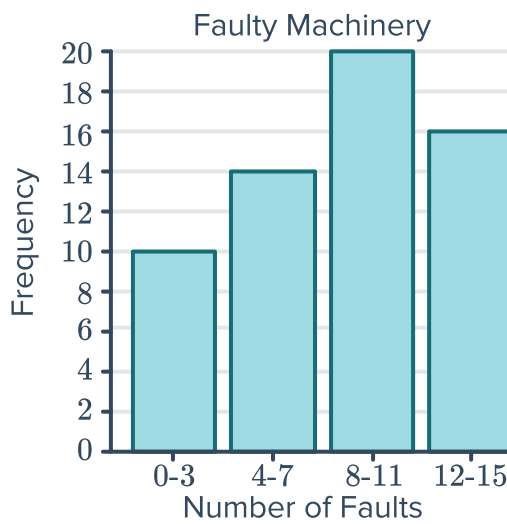
The areas are the same.



21

a

b 0

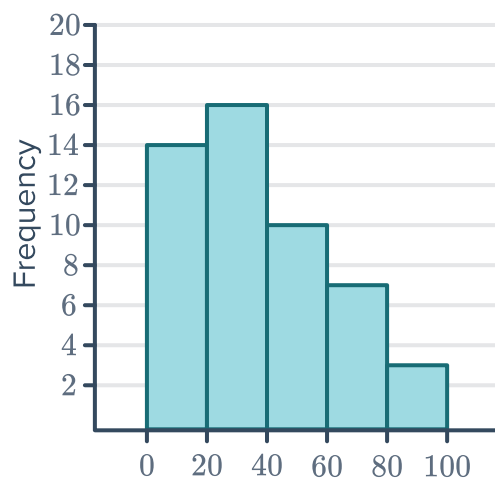


22

a

b

i No ii Yes iii No iv No



23 a

i $0 \leq \text{rate} < 5$

iii $10 \leq \text{rate} < 15$

b 3



ii $5 \leq \text{rate} < 10$

iv $15 \leq \text{rate} < 20$

24

a 140.9 cents

b 11 days

25

a 25 students

b 13 students

c 16%

26

a

Height (cm)	Frequency
$95 \leq h < 100$	1
$100 \leq h < 105$	3
$105 \leq h < 110$	2
$110 \leq h < 115$	12
$115 \leq h < 120$	19
$120 \leq h < 125$	15
$125 \leq h < 130$	4
$130 \leq h < 135$	4
Total:	60

b 31.7%

27

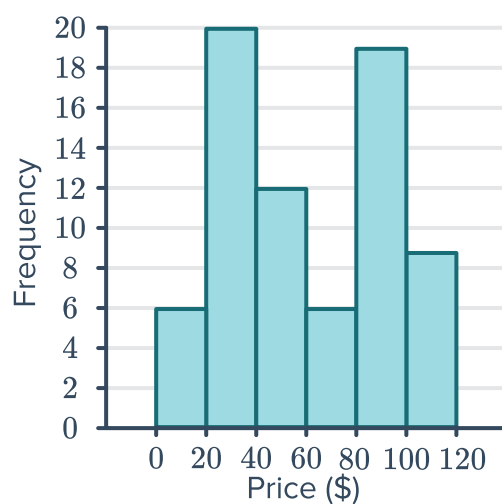
a

Speed (km/h)	Frequency
$110 \leq s < 120$	4
$120 \leq s < 130$	12
$130 \leq s < 140$	8
$140 \leq s < 150$	5
$150 \leq s < 160$	2
Total:	31

b 51.6%

28

a



b 20 – 39 and 80 – 99

c The data shows that there are two main price ranges of books that were recently bought by university students: \$20 to \$39, and \$80 to \$99.

29

a

Battery life (minutes)	Frequency
$400 \leq x < 500$	3
$500 \leq x < 600$	9
$600 \leq x < 700$	13
$700 \leq x < 800$	32
$800 \leq x < 900$	22
$900 \leq x < 1000$	27
$1000 \leq x < 1100$	6
$1100 \leq x < 1200$	4
$1200 \leq x < 1300$	1
Total :	117

b 89.7%

c 9.4%

30



- a 35
- b $60 \text{ mm} \leq x < 80 \text{ mm}$
- c Yes
- d No. The table shows ranges of lengths and not individual leaves. We do not know if any of the leaves in the range $0 \leq x < 20$ are less than 5 mm.